Analytical Cume

January 18, 2008

Analytical Figures of Merit.

Perhaps the most important analytical figure of merit is that of signal to noise ratio (S:N). A measurement with a S:N of 100:1 can be thought of as signal S of 100, with an uncertainty, e, of 1. That is the measurement of is:

$S \pm e = 100 \pm 1$

1. (20 points) Use the rules of propagation of uncertainty to show that if you add two such signals, the resulting signal to noise is 141:1.

   In case you forgot, the form is:

E1 ± e1
E2 ± e2
E3 ± e3
.
.
.
<u>+ En±en</u>
Es ± es Where es is given by: $es = \sqrt{e1^2 + e2^2 + e3^2 + \bullet\bullet\bullet + en^2}$

Es is the value of the total sum, es is the uncertainty in the sum.

En is the nth number in the sum, with it's associated uncertainty en
In this example the uncertainties (e) are the same as the noise (N).

2. (20 points) Show that summing n measurements increases the S:N by a factor of $\sqrt{n}$ compared with the value for one measurement.

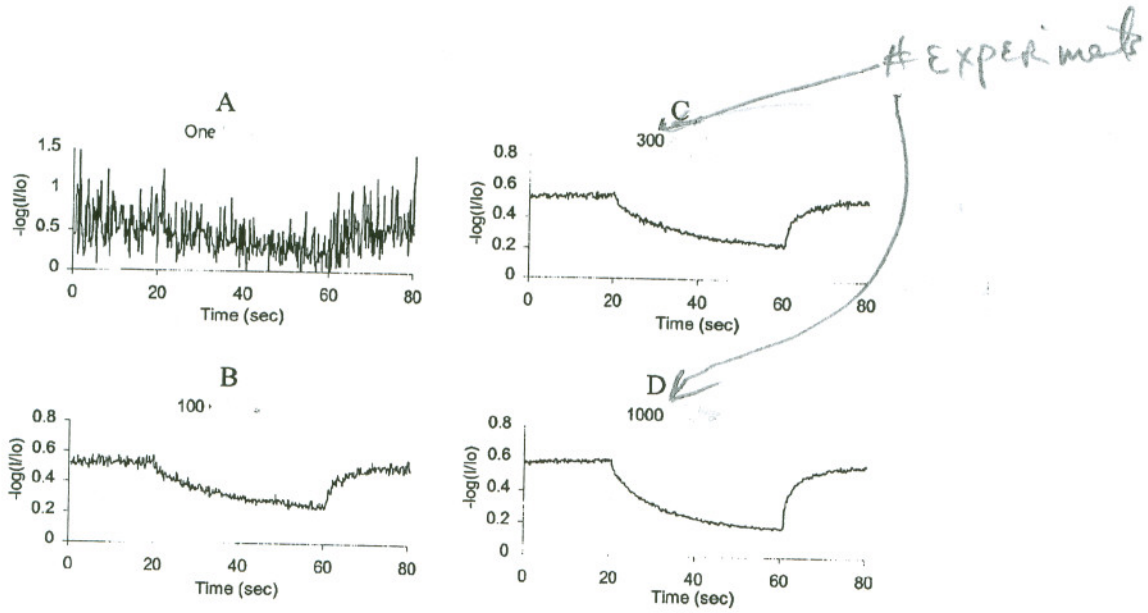
3. (20 points) If the S:N is 60 from summing 1000 experiments predict the S:N by summing the following number of experiments:
   A) 300
   B) 100
   C) 1

4. (10 points) For signal summing to work, what experimental requirements must be met?

5. (10 points) Another figure of merit is limit of detection. Limit of detection is defined as the concentration needed to cause a S:N of at minimum 3:1. Questions

1-4 are about only one concentration, and in this case 1.0×10^{-5} M. How many experiments need to be averaged to achieve a detection limit of 1.0×10^{-5} M?

6. (20 points) How does the limit of detection depend upon the number of experiments averaged?



Staterbeck, Ridgway, Seliskar, Heineman, Anal. CHem. 1999, 71, 1196-1203